

Integrated Photovoltaic Energy Harvesting System with Charge Pump Conversion and LDO Regulation

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Integrated Photovoltaic Energy Harvesting System with Charge Pump Conversion and LDO Regulation

*Project report submitted to the
SRM University - AP, Andhra Pradesh
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of

Bachelor of Technology

by

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APRIL, 2026

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List of Abbreviations

MPPT	Maximum Power Point Theorem
VCO	Voltage Controlled Oscillator
PV Cell	Photovoltaic Cell
LDO	Low-Dropout Regulator
BGR	Bandgap Reference
ML	Machine Learning
PMOS	P-channel Metal Oxide Semiconductor
NMOS	N-channel Metal Oxide Semiconductor
CTAT	Complementary To Absolute Temperature
PTAT	Proportional To Absolute Temperature

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Abstract

With the quick development of low-power microelectronics and edge devices that can sustain themselves, it is necessary to have very efficient power management. In this thesis, we design and evaluate the performance of a full-fledged Power Management Unit (PMU) designed to harvest power using photovoltaics and provide regulated voltages. Our proposed design uses an intelligent harvesting front-end employing an MPPT controller based on machine learning techniques to extract maximum power from photovoltaics depending on the ambient conditions. Finally, the generated power undergoes processing via a Voltage Controlled Oscillator (VCO) and a specially designed charge pump to increase voltages. In order to provide a very stable and scaleable power supply voltage for dynamic loads, a Low dropout regulator (LDO) can be incorporated into the chip. The LDO architecture includes a well-designed pass transistor component, an amplifier with a large gain, and a digitally programmable feedback circuit that enables the scaling of the output voltage. In addition, in order to ensure precision in the overall circuit, a stable BGR component, consisting of a PMOS-based error amplifier and well-designed resistor ratio components, can be included in the design, providing a stable reference voltage irrespective of temperature variations and power supply variations.

Keywords: Charge pump, Voltage controlled oscillator, MPPT, Low dropout regulator, BGR.

CHAPTER 1

Introduction

1.1 Introduction

Fast growth and innovation in the realm of microelectronic devices and computing platforms have resulted in the necessity for more efficient ways of power delivery and management for them. Traditional methods that involve utilization of batteries suffer from such drawbacks as limited lifetimes and scalability problems. Hence, energy harvesting technologies have gained popularity for providing the ability to perform continuously and sustainably. Photovoltaics play an essential role in this case.

However, since the obtained amount of power from photovoltaics depends heavily on such factors as light intensity and ambient temperature, the ability to efficiently harvest and regulate it becomes critical for such energy sources. In this research paper, an effort was made to develop a complete system for performing power management and harvesting from the photovoltaic cell with voltage booster and regulator implemented as part of a PMU. For maximizing the performance of the system, the Maximum Power Point Tracking (MPPT) algorithm was applied in the design of the intelligent front-end. Energy was further regulated with the help of the Voltage Controlled Oscillator (VCO) and charge pump.

In order to guarantee dependable functioning of downstream components irrespective of changes in the load, Low-Drop-Out Regulator (LDO) has been designed to deliver a constant scalable voltage output. This regulator uses high gain amplifier, pass transistors, and programmable feedback path to enable output voltage scaling. Furthermore, Bandgap Reference (BGR) block is used in the system to produce constant voltage regardless of temperature changes and input voltage.

This design caters to the limitations encountered in energy harvesting, making it ideal for future generation devices.

1.2 Literature Survey

The growing demand for secure and energy-efficient IoT systems has led to significant research in hardware security and power management. Ram *et al.* proposed a configurable ring oscillator-based PUF for secure and reliable IC identification in constrained environments [1].

In energy harvesting, Ranjith *et al.* combined machine learning with P&O-based MPPT to improve tracking under dynamic conditions [2]. Devaraj *et al.* developed a switched-capacitor multi-input harvester integrating solar and piezo sources with high efficiency [3]. Khanam *et al.* further enhanced MPPT performance using machine learning-based adaptive techniques [4].

For efficient power conversion, Zhang *et al.* introduced a modified Dickson charge pump achieving higher voltage gain and efficiency [5]. Tabrizi *et al.* designed an autonomous MPPT-enabled interface for thermoelectric generators [6]. Pinninti *et al.* proposed a digitally controlled reconfigurable charge pump for adaptive solar harvesting [7].

At the system level, Ram *et al.* developed Eternal-Thing 2.0, a ripple-less and Trojan-resilient solar harvesting system [8], followed by Eternal-Thing 3.0, a mixed-mode SoC for scalable IoT energy solutions [9].

Voltage regulation is critical for such systems. Wu *et al.* proposed a programmable output LDO for flexible voltage scaling [10], while Milliken *et al.* demonstrated a fully integrated CMOS LDO with improved stability [11].

Ultra-low-power designs have gained focus, with Rashidian *et al.* presenting a nano-watt BGR with curvature compensation [12], and Abd Aziz *et al.* developing an efficient LDO in 0.18 μm CMOS [13].

In BGR design, Lee *et al.* introduced a sub- μW circuit with inherent curvature compensation [14], while Alam provided foundational amplifier-based BGR design approaches [15].

For IoT applications, Kim *et al.* proposed an ultra-low quiescent current LDO with adaptive modes [16]. Additional low-power and capacitor-less LDO techniques were explored in [17–19].

Low-voltage BGR evolution includes Banba *et al.*'s sub-1V CMOS design [20], Pereira's ultra-low-power capless LDO [21], and Dalal *et al.*'s sub-1V BGR implementations [22].

Fundamental LDO concepts were detailed by Day [23], while Lin and Liang demonstrated subthreshold BGR operation for improved efficiency [24]. Hu *et al.* proposed a high-performance Brokaw BGR with enhanced stability [25].

1.3 Gap Analysis

Current research efforts mainly aim to improve individual modules including MPPT, charge pumps, LDOs, and bandgap references without giving much thought to the integration process of such components within an entire energy harvesting system. Although machine learning algorithms have proven useful in achieving optimal energy harvesting, the effect of such methods on downstream analog blocks is still not investigated.

Furthermore, the design of charge pumps aims at improving voltage boost and efficiency but does not take the effect on the regulation stage due to ripple generation and instability into account.

In addition, almost all existing LDO architectures aim at minimizing their performance metrics (dropout, PSRR) while ignoring their adaptability under dynamic loading conditions. Moreover, integrating stable reference circuits in such energy constrained, low-voltage systems is overlooked.

The current research attempts to integrate a full energy harvesting system that includes the MPPT algorithm, voltage boost through charge pump architecture and LDO architecture along with a stable reference circuit.

1.4 Motivation

The increasing need for autonomous electronic systems in IoT and edge computing systems demands more efficient and practical solutions to energy harvesting. Batteries cannot be relied upon as the main source of energy due to their short-

comings when it comes to scalability and longevity of operation. Thus, energy harvesting technology emerges as a more practical solution to powering such devices.

Nevertheless, the energy harvested using the energy sources is limited and unreliable since it varies depending on the environmental factors. It, therefore, necessitates the efficient extraction, boosting, and regulation of the energy. Most of the existing researches focus on tackling each step separately without ensuring full integration of the processes. Thus, it becomes necessary to create a solution to ensure compatibility among different processes in order to create an efficient and stable system.

1.5 Problem Statement

As regards the sources of energy harvesting, solar cells produce low, unpredictable, and unsteady energy, which makes it hard to power electronic devices directly. It is, therefore, important to consider how efficiently energy can be harvested, converted, and regulated from such sources. The development of an efficient energy harvesting system is hence essential for optimal power extraction and boosting, as well as for maintaining stable voltage levels.

1.6 A Layout of the Report

The report is structured in the following way:

- Chapter 1 contains Introduction
- Chapter 2 contains Background and Theoretical Foundations
- Chapter 3 contains Proposed System Design and Implementation
- Chapter 4 contains Programmable Low Dropout Regulator
- Chapter 5 contains System Architecture
- Chapter 6 contains Result and Analysis



Background and Theoretical Foundations

2.1 Introduction

Photovoltaic (PV) systems are considered effective methods for energy harvesting through the simple and robust way of transforming solar energy directly into electric energy. Such systems find a wide area of application as power sources in IoT applications, wireless sensor nodes, and portable electronic devices.

Nevertheless, photovoltaic systems have many disadvantages. First of all, there are no linear dependencies between the current and voltage in photovoltaic cells. Moreover, these parameters strongly depend on external factors, such as solar irradiance and temperature. Under certain conditions, the PV cell has a special point where the output power reaches its maximum value, which is called Maximum Power Point (MPP). As MPP changes depending on environmental conditions, it is vital to control the system in order to track this point.

Secondly, the output voltage of the PV cell is usually insufficient for energy conversion. Hence, it requires a special circuit to increase and stabilize the output voltage. In this study, the authors apply a charge-pump based DC-DC converter together with an MPPT algorithm based on machine learning.

An LDO is one of the vital blocks in our energy harvesting system. The basic idea behind LDO lies in the generation of constant and noise-free output voltage regardless of the fluctuations occurring at input and loads side. It includes a feedback control circuit consisting of a pass transistor, an error amplifier, and a feedback network to control the output voltage by keeping minimum voltage differential between input and output voltage. Due to simplicity, lower noise levels, and faster operations, the LDO finds its application in analog and mixed-signal circuits, particularly those which are energy-efficient.

Another important block is the Bandgap Reference (BGR) block that helps generate a precise reference voltage which is almost independent of changes in temperature and supply level. This precision can be achieved because of combining two different voltages with opposite temperature coefficients.

The BGR is the vital part in LDO since it generates the necessary reference voltage for the LDO to function efficiently. In this way, we have a complete voltage

regulator consisting of BGR and LDO, where the former generates reference voltage while the latter generates desired output by regulating the voltage to match the desired reference level.

This chapter provides the theoretical background regarding photovoltaic cells' operation and principles of PV systems, as well as the theory behind power extraction technologies followed by LDO which works with the help of BGR.

2.2 Photovoltaic Cell Fundamentals

Photovoltaic cells are semiconductor-based devices that harness the photovoltaic effect for converting light energy into electricity. They generally consist of a p-n junction that produces electron-hole pairs upon illumination by photons. The material's internal electric field separates the charges, and a circuit completes the current.

The nonlinear I-V and P-V curves act as the foundation for a PV cell's operation.

2.2.0.1 I-V Characteristics

The current-voltage relationship of a PV cell is given by:

$$I = I_{ph} - I_0 \left(e^{\frac{V+IR_s}{nV_T}} - 1 \right) - \frac{V + IR_s}{R_{sh}} \quad (2.1)$$

This expression comprises three major elements:

- Photocurrent (I_{ph}): Resulting from incident solar.
- Diode current: Depicting recombination losses at the p-n junction.
- Shunt current: Depicting leakage current due to defects.

The nonlinear exponential term is responsible for making the PV characteristic very sensitive to voltage variations.

2.2.0.2 Thermal Voltage

The thermal voltage is defined as:

$$V_t = \frac{kT}{q} \quad (2.2)$$

A rise in temperature raises the thermal voltage, which influences the diode current and lowers the PV cell's total output voltage.

2.2.0.3 Power Characteristics and Maximum Power Point

The output power of the PV cell is given by:

$$P = V \cdot I \quad (2.3)$$

The power-voltage (P–V) curve has a unique peak known as the Maximum Power Point (MPP). At this point:

$$\frac{dP}{dV} = 0 \quad (2.4)$$

Operating at MPP ensures maximum energy extraction from the PV cell.

2.2.0.4 Effect of Irradiance and Temperature

The PV characteristics vary with environmental conditions:

- **Irradiance (G):** Directly proportional to photocurrent, Higher irradiance increases output current and power.
- **Temperature (T):** Increases diode conduction, Reduces open-circuit voltage, and Slightly increases the current.

Due to these variations, the MPP shifts continuously, making static operation inefficient.

2.3 PV Cell Modeling

Modelling of PV cells becomes very important in understanding their behaviour under various climatic conditions and in designing effective control algorithms like MPPT. As the behaviour of PV cells is highly nonlinear, equivalent circuit modelling is done for their analysis.

The most common equivalent circuit model is the single-diode model. This model offers a perfect balance between accuracy and complexity.

2.3.0.1 Single-Diode Model

The components included in a single-diode model of a PV cell include:

- Current source denoting the photo current due to sunlight
- Diode denoting the p-n junction properties.
- The series resistor, denoted as R_s , which denotes the resistance inside the PV cell
- Shunt resistor denoted by R_{sh}

It includes both ideal and non-ideal factors of a PV cell.

2.4 Maximum Power Point Tracking (MPPT)

Maximum Power Point Tracking (MPPT) refers to an approach that involves the tracking of the maximum amount of energy output in photovoltaic systems, depending on the changes in various environmental factors. The nonlinear property of PV characteristics necessitates the use of dynamic adjustment of the operating point of the system to maximize the efficiency of energy production.

Since the MPP is not constant and changes based on variations in environmental factors, a control algorithm needs to be implemented to track and sustain this point.

Maximum Power Point Tracking (MPPT) involves controlling either voltage or current by employing a power converter in photovoltaic systems. The adjustment of parameters like duty cycles allows the load resistance to change the operating point of the PV cell.

Several conventional MPPT techniques have been developed, but the most common include:

Perturb and Observe (P and O): This technique consists of periodically changing the working voltage and observing how it affects the generated power. Depending on whether there is an increase or decrease in power, the system determines the direction in which further modification should be carried out. Although simple to implement, this approach has drawbacks, including oscillation near the peak value of power and slow responses to sudden environmental changes.

Even though these conventional algorithms work effectively, they suffer from several disadvantages, such as low speed of convergence, oscillation during steady state, and noise sensitivity.

A more modern alternative to overcoming the shortcomings of conventional algorithms is using a machine learning (ML)-based MPPT algorithm. This approach does not involve continuous search for the optimum point; rather, the algorithm predicts the optimal point based on the environment data.

Environmental data, such as radiation intensity and temperature, are input into the ML model. The model predicts the voltage value at which the PV system operates to generate maximum power.

2.5 DC-DC Conversion and Charge Pump Fundamentals

In most practical PV systems, however, the voltage produced from the PV cells is insufficient to directly drive electronic circuits. For that reason, some stage of power conversion becomes necessary to boost the voltage level.

There are several power conversions techniques that are utilized in such appli-

cations. In this work, a DC-DC converter based on the charge pump architecture was adopted owing to its suitability for integrated circuit realization.

Charge pump refers to a particular type of DC-DC converters. It is characterized by the absence of bulky and costly inductors as compared to conventional architectures. The conversion process relies solely on capacitors that get reconfigured according to the switching signals of a clock.

Essentially, the charge pump works through repeated charges and discharges cycles in which energy gets transferred using capacitors to achieve voltage boosting. For a voltage doubler, theoretically, the output voltage should double the input, but practically, it falls short of doubling it.

The charge pump in this project is responsible for the voltage boosting stage after MPPT controller has been employed. The switching signals produced by the MPPT have direct impact on how the pump works in terms of switching actions.

Through switching actions, the pump controls the effective load that sees the PV cell. Through that, the MPPT controller manages to control the operating point of PV system

2.6 LDO Fundamentals

LDO or Low Drop-Out regulator refers to a linear regulator designed to produce a stable output voltage regardless of how little is the difference between the input and output voltages. In such a case, it is considered a better choice for low power systems since it ensures effective use of the limited supply voltage. Some other features include low output ripple and fast response time, making them suitable for analog and mixed signal components.

The term dropout voltage refers to the minimum difference between the input and output voltages necessary for the LDO to function properly. Dropout voltages are usually 30mV – 300mV.

$$V_{\text{dropout}} = V_{\text{in}} - V_{\text{out}} \quad (2.5)$$

The main components of LDO include a pass transistor, error amplifier, feedback circuit, and reference voltage generation. This reference voltage generation is commonly called the Bandgap Reference Circuit.

The output voltage is sensed using the resistor network for feedback purposes and compared against the reference voltage. An error voltage, as generated from the differential comparison of these two voltages, is amplified using an error amplifier and used to control the gate of the pass transistor.

$$V_{\text{out}} = V_{\text{ref}} \left(1 + \frac{R_1}{R_2} \right) \quad (2.6)$$

2.6.1 Pass Element

The function of the pass element within the LDO is to control the current flow from the input supply to the output side and regulate the output voltage. It acts as a voltage-controlled current source, whose gate voltage can be adjusted by the error amplifier to ensure a constant output voltage level.

Pass element performance affects important parameters, including the dropout voltage, load regulation, and efficiency. The pass element should always have sufficient size to provide the necessary current for the load and minimize the resistance across the pass element.

2.6.2 Error Amplifier

The error amplifier is one of the most important parts in the feedback network of an LDO, where it compares the divided output voltage and the reference voltage to produce an input signal to the pass device. This process produces an error voltage which then undergoes amplification using the error amplifier. It is important for the amplifier to have high gain to reduce errors.

2.6.3 Feedback Network

The feedback circuit is made up of a voltage divider made of two resistors that measure the output voltage and deliver a proportion of it to the error amplifier.

This creates a closed loop where the output voltage can be kept stable despite changes in the input voltage and load current. The resistance ratio sets the regulated output voltage.

2.7 Bandgap Reference Fundamentals

Band Gap Reference (BGR) is described as an electronic circuit that generates a constant voltage reference independent of temperature variations. This is often applied in various analog and mixed-signal applications, which include LDOs. The function of the band gap reference is achieved through the use of two voltage generators with opposite temperature coefficients; these include CTAT and PTAT.

Traditional bandgap voltage generators provide voltages close to 1.2 V. Such voltages are dependent on intrinsic silicon properties and temperature-dependent voltage generators. The collector-base voltage V_{BE} of a bipolar junction transistor (BJT) generator has a CTAT characteristic and usually ranges from 0.6 to 0.7 V under room conditions, depending on rising temperature. On the other hand, the differential voltage generator ΔV_{BE} between the collector base voltages of two BJTs with differing current densities has a PTAT characteristic.



MPPT and Charge Pump System Design

3.1 Introduction

The design and implementation of the suggested PV energy harvesting system is outlined in this chapter. The system includes an MPPT controller using machine learning, along with a VCO and a charge pump.

In the suggested design, the primary goal is to harvest the energy from the photovoltaic source and convert it into a required voltage level. While the MPPT controller suggests an optimal operating point for the operation, the VCO along with the charge pump makes sure that the switching takes place.

3.2 Overall System Architecture

The entire system contains the following main blocks:

- Solar PV Cell
- MPPT controller using Machine Learning
- Voltage Controlled Oscillator (VCO)
- Charge Pump (Voltage Doubler)

The solar PV cell converts light energy to electrical energy from the environment. The MPPT controller takes input data regarding sensing parameters and provides a duty cycle. The duty cycle is converted to switching signals by using the VCO. The generated clock pulses operate the charge pump, which doubles the output voltage.

This system acts as an energy harvesting converter circuit.

3.3 Photovoltaic Cell Design

A Verilog-A model was developed for the photovoltaic cell using a model that depends on a single diode equivalent model. Some of the features that have been captured include:

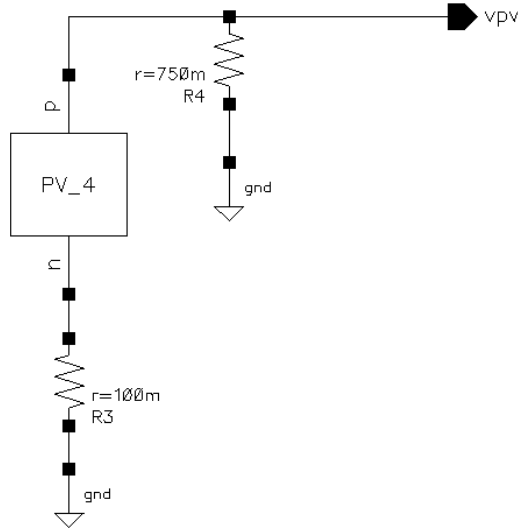


Figure 3.1: Photovoltaic Cell Model

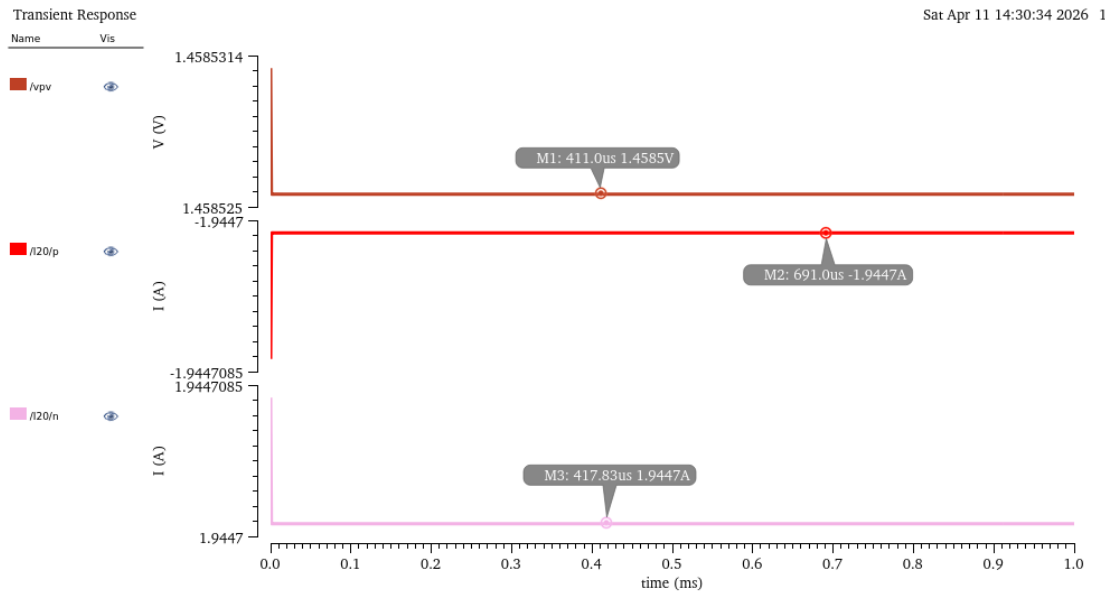


Figure 3.2: Output Characteristics of Photovoltaic Cell

- Irradiance dependence
- Variation due to temperature
- Resistive losses
- Current-voltage nonlinearity

The PV cell generates a voltage and current output, which serves as the input to the MPPT control algorithm. The sensing process has been implemented to obtain the PV current through a small sensing resistor.

The schematic representation of the PV cell model implemented in the Cadence environment is shown in Figure 3.1. The corresponding output characteristics, illustrating the voltage and current behavior of the PV cell under simulation conditions, are presented in Figure 3.2.

3.3.1 Dataset and Model Development

Table 3.1: Sample Photovoltaic Dataset

G (W/m ²)	T (°C)	V_{mp} (V)	I_{mp} (A)	P_{mp} (W)	V_{oc} (V)	I_{sc} (A)
100	10	1.7598	0.0454	0.0799	2.09	0.0493
100	32.1	1.6188	0.0460	0.0745	1.96	0.0504
100	54.2	1.4836	0.0465	0.0690	1.82	0.0515
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1000	72.6	1.35609	0.47267	0.64098	1.7144	0.5238
1000	76.3	1.33515	0.47263	0.63103	1.6922	0.52565
1000	80	1.31262	0.47313	0.62104	1.67	0.5275

A dataset was generated to represent the behaviour of the photovoltaic system under varying irradiance and temperature conditions. The dataset includes parameters such as PV voltage (V_{mp}), current (I_{mp}), and power (P_{mp}), along with open-circuit voltage (V_{oc}) and short-circuit current (I_{sc}).

The data shows that as temperature increases, the maximum power point voltage decreases, while increasing irradiance leads to higher current and power output. This reflects the practical characteristics of photovoltaic systems.

The dataset was processed using Python, where multiple regression models were evaluated to estimate the maximum power point voltage as a function of irradiance and temperature. The best-performing model coefficients were then implemented in Verilog-A for circuit-level simulation.

3.3.2 Model Performance and Selection

Several machine learning models were evaluated to determine the most suitable approach for MPPT prediction. Performance was assessed using metrics such as R^2 , Root Mean Square Error (RMSE), and Mean Absolute Error (MAE).

Among the evaluated models, polynomial regression (degree 2) and gradient boosting showed the highest accuracy with minimal error. However, linear regression was selected for implementation due to its simplicity, low computational complexity, and ease of integration into Verilog-A.

This selection ensures a balance between prediction accuracy and hardware feasibility.

3.4 Voltage Controlled Oscillator (VCO)

The VCO is responsible for converting the analog duty cycle signal into switching clock signals necessary to operate the charge pump.

The frequency of oscillation depends on the input duty signal. The higher or lower the input duty cycle, the higher or lower will be the frequency of the clock signals.

The VCO generates the following clock signals:

- Clock signal 1 (CLK1)
- Clock signal 2 (CLK2)

These are the clock signals that are needed for controlling alternating switching modes within the charge pump.

3.5 Charge Pump Design

The charge pump is implemented as a voltage doubler circuit to boost the output voltage from the PV source.

3.5.1 Operating Principle

The working of the charge pump can be explained by the movement of charges from one capacitor to another using switching action.

This ensures efficient energy transfer through the use of complementary signals.

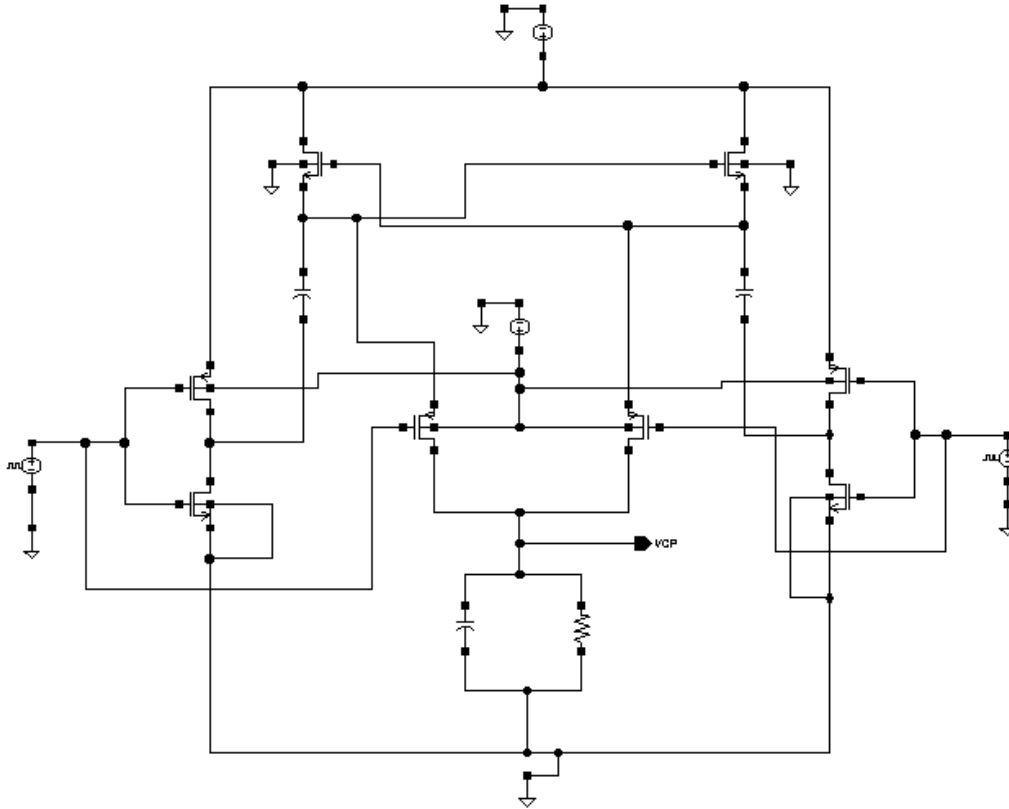


Figure 3.3: Voltage Doubler Circuit

3.5.2 Circuit Implementation

The following parts make up the charge pump:

- Transistors for switching
- Capacitors
- Controlled signals from the clock

A design for the charge pump that works with the output of the VCO clock signals can be made.

3.5.3 Output Characteristics

The voltage output from the charge pump is greater than that of the input from the PV panel. The voltage output is used to drive low-voltage electronics.

The efficiency of the charge pump is dependent on switching rate, loading and capacitance. Figure 3.4 shows the transient response of the charge pump,

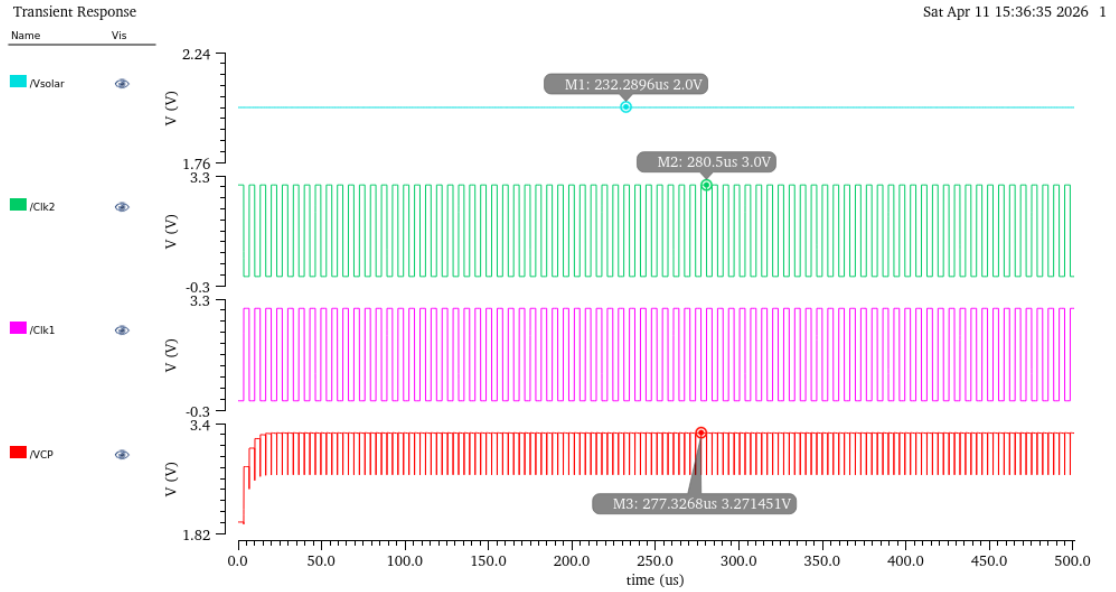


Figure 3.4: Output Characteristics of Voltage Doubler Circuit

where the input voltage of approximately 2 V is boosted to around 3.27 V using complementary clock signals.

3.6 Integration of System Components

All blocks are connected to make a system that can harvest energy.

- PV cell is used as an energy source
- MPPT controller receives signals and provides the duty cycle
- VCO converts duty to clock
- Charge pump increases voltage

This interplay of blocks allows the circuit to operate continuously and harvest energy.



Programmable Low Dropout Regulator

4.1 Introduction

The modern system-on-chip (SoC) designs, especially in the context of IoT devices operating with battery constraints, have necessitated a requirement for dynamic power management. This is achieved by adopting a programmable design approach for the LDO regulator. Traditional LDO regulators employ a constant feedback fraction in order to regulate only one output voltage. However, in the present design, a digital programmable design approach has been used to achieve a range of output voltages as well as a power-down mode by using a 2-bit digital input (S_0, S_1).

4.2 Design Methodology and Block Diagram

Top-down design methodology is used in designing the circuitry of the LDO where one starts from the output voltage requirement and stability considerations, then proceeds with the design of each block in sequence. A reference voltage (V_{ref}) is first chosen via bandgap reference which acts as the input for the error amplifier. Designing of the feedback network comprising of resistors R_1 and R_2 helps define the required output voltage based on the feedback relationship. Resistors R_3, R_4 , along with some control switches S_0 and S_1 , can be included to achieve programmable or selectable modes within the feedback path.

The error amplifier compares the feedback voltage with V_{ref} and gives an output which is buffered before driving the gate terminal of the pass transistor. The buffering process ensures that the high gain amplifier stage is isolated and that sufficient drive current is provided for charging the gate capacitance of the pass transistor. An appropriate sizing for the pass transistor is done considering the necessary current delivery while keeping the drop out voltage at a minimum. Load elements R_L, C_L are accounted for in the design process.

The design starts with the choice the pass transistor element according to the load current and dropout requirements. After choosing the proper transistor, an error amplifier having sufficient gain is designed for achieving high accuracy.

Subsequently, the feedback network is designed for achieving the required output voltage and power dissipation level. A stable bandgap reference source is also included for providing a stable reference voltage.

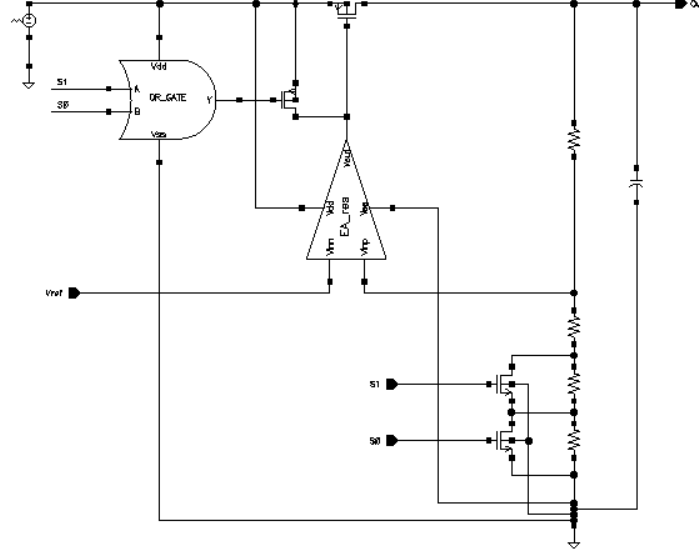


Figure 4.1: Schematic diagram of the circuit

Using programmable devices (M_0, M_1, S_0, S_1) provides flexibility in terms of performance, allowing adjustments in output voltage levels or ensuring stability under different load conditions.

Output Voltage is given as

$$V_{\text{out}} = V_{\text{ref}} \left(1 + \frac{R_1}{R_{\text{effective}}} \right) \quad (4.1)$$

Where $R_{\text{effective}}$ is the effective resistance based on S_0 and S_1

4.3 Pass Transistor Element

The pass transistor plays an important role in the working of the LDO by controlling the current flow from the supply to the load. The use of the PMOS transistor in the design is justified by the fact that the PMOS transistor can control the output voltage in case of low dropout. The reason why the PMOS transistor

can be used in the low dropout regulator is that the gate voltage required for its operation should not exceed the supply voltage.

The PMOS transistor mainly operates in the saturation region because regulation of the output voltage depends on the mode. Therefore, the output voltage is regulated through modulation of the gate-source voltage of the PMOS transistor.

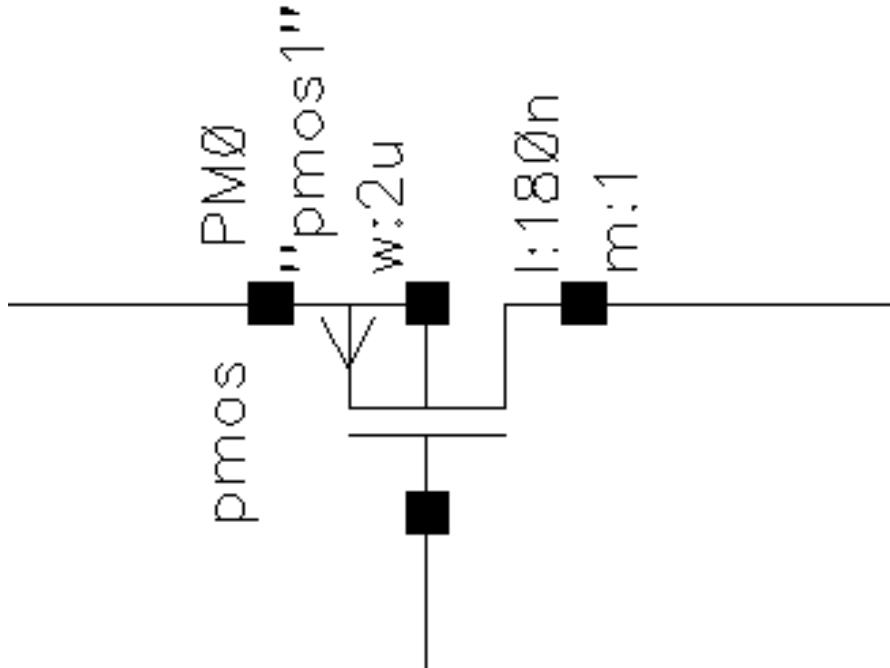


Figure 4.2: PMOS pass transistor

The current through the PMOS pass transistor is given by

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_{TH}|)^2 \quad (4.2)$$

Where

μ_p - hole mobility

C_{ox} - oxide capacitance

W/L - transistor aspect ratio.

When the input voltage increases or decreases, the pass transistor increases or decreases its V_{DS} respectively to keep the output constant with respect to the input.

4.4 Error Amplifier

Error amplifier is part and parcel of LDO circuitry that acts as an amplifier of difference between feedback voltage and reference voltage. The gate of pass PMOS device is controlled by error amplifier to set the necessary operating point in the DC domain. Moreover, error amplifier determines the output voltage, which is compared with reference voltage created by bandgap reference (BGR).

4.4.1 Topology

For this purpose, this work utilizes a two-stage CMOS operational amplifier with an NMOS differential input stage and a common source gain stage. Such an architecture provides the required open-loop gain and output swing, enabling sufficient gate drive of the PMOS pass transistor.

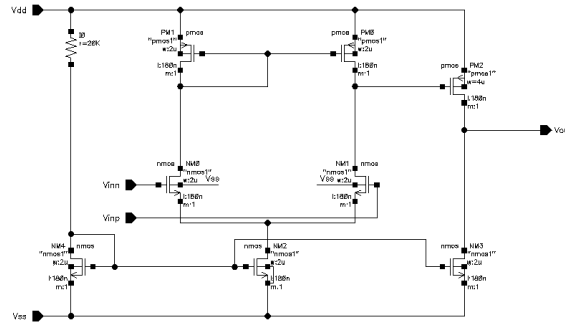


Figure 4.3: Error Amplifier

In this case, the first stage involves the use of an NMOS differential input stage and a PMOS current mirror load. The main function of this stage is to provide the initial amplification of the differential input voltages and generate differential currents. This design employs an NMOS input stage because it offers a larger transconductance for every unit of biasing current.

4.4.2 Small Signal Analysis

The overall voltage gain of the error amplifier can be approximated as

$$A_v = (g_{m1}r_{o1}) \cdot (g_{m2}r_{o2}) \quad (4.3)$$

Where g_{m1} , g_{m2} are the transconductances of the input NMOS and common source NMOS of the second gain stage respectively, similarly r_{o1} , r_{o2} are output resistances of those transistors

Proper biasing of the amplifier is necessary for proper gain and linearity to be achieved, and all transistors should work in saturation mode. The input signal's common mode voltage depends on the performance of the feedback system, and this must be kept within the input common-mode voltage range of the differential amplifier. In a similar vein, the reference voltage must be within the input common-mode voltage range as well.

Below are the Frequency Analysis and CMRR curve of the error amplifier.

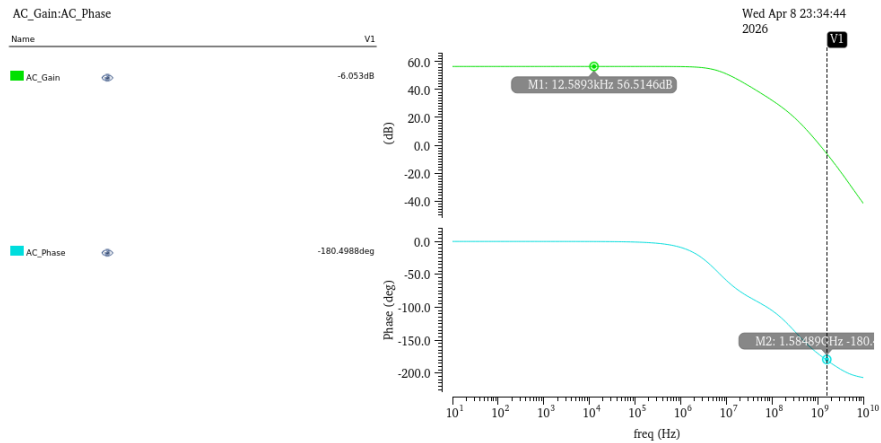


Figure 4.4: Frequency Response of the Error Amplifier

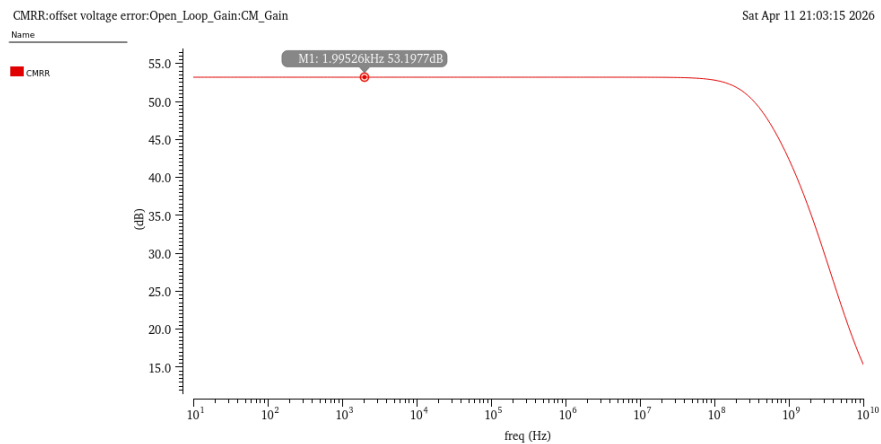


Figure 4.5: Common-Mode Rejection Ratio

4.5 Feedback Network

In an LDO, the feedback network is tasked with sensing the output voltage and scaling it down before presenting it to the error amplifier in comparison to the reference voltage. In our particular case of programmable output configuration, a resistor network made up of four resistors, R_1 , R_2 , R_3 and R_4 are employed for producing the feedback voltage. This voltage appears at the non-inverting terminal of the error amplifier while the reference voltage goes into the inverting input.

4.5.1 Topology

In the LDO, the feedback system consists of a digitally programmable resistor divider. In this configuration, the output voltage V_O can be controlled by varying the resistance of the lower resistor string through signals S_1 and S_0 as shown in figure 4.4 below.

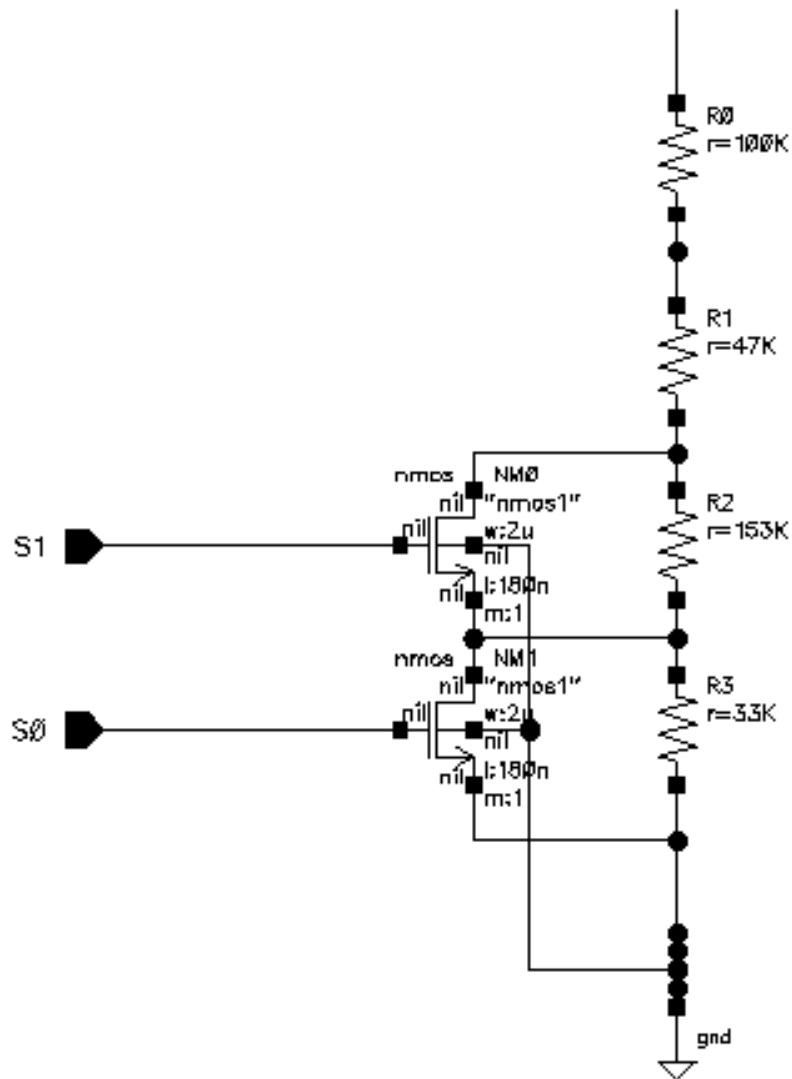


Figure 4.6: Feedback Resistors

4.5.2 Programming Logic

The NMOS transistors M_1 and M_0 function as switches to create a short-circuit

Table 4.1: Comparison of Output Voltage

S_1	S_0	V_{out}
0	0	0
0	1	$V_{ref} \left(1 + \frac{R_1}{R_2 + R_3} \right)$
1	0	$V_{ref} \left(1 + \frac{R_1}{R_2 + R_4} \right)$
1	1	$V_{ref} \left(1 + \frac{R_1}{R_2} \right)$

for certain resistors within the series, thus changing r_{bottom} . Switch M_1 : When S_1 is high, M_1 turns on and shorts R_3 . Switch M_0 : When S_0 is high, M_0 turns on and shorts R_4 .

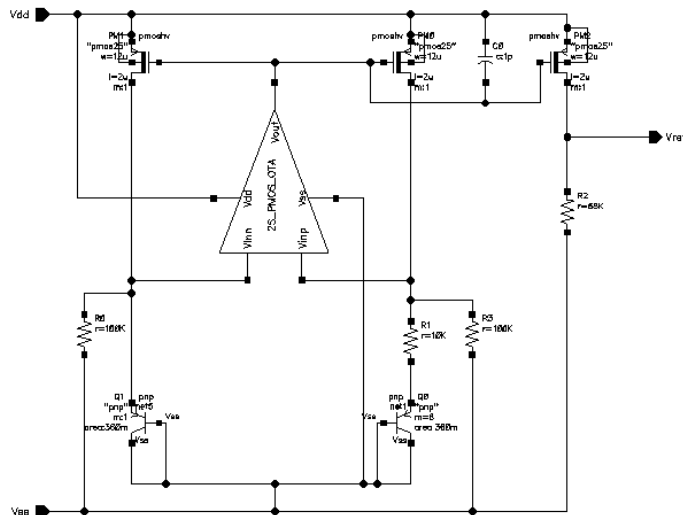
4.6 Bandgap Reference

The Band Gap Reference (BGR) forms a temperature-independent reference voltage source for the low-dropout regulator (LDO). In this design, a current-mode BGR is used, where a reference voltage is first generated from a current whose value is independent of temperature, and then the temperature-independent current is converted to a reference voltage. The above process will improve the supply rejection of the circuit and integrate with current-mode biasing in the LDO.

The function of the current-mode band gap reference is based on a combination of Proportional To Absolute Temperature (PTAT) and Complementary To Absolute Temperature (CTAT). The value of the CTAT voltage is proportional to the base-emitter voltage (V_{BE}) of the bipolar junction transistor, whereby its value becomes lower with the increase in temperature. However, the value of the PTAT voltage is equal to the difference in base-emitter voltage of two transistors.

4.6.1 Topology

We are going to incorporate the Banba Topology. Banba BGR is a commonly adopted low-voltage BGR design which can perform well within limited power



The core idea of the Banba circuit is that it generates a temperature-independent current by adding a CTAT current that comes from the base-emitter voltage V_{BE} of a bipolar transistor and a PTAT current that originates from the difference between base-emitter voltages of two bipolar transistors (ΔV_{BE}), which operate at different current densities.

$$I_{ref} = I_{PTAT} + I_{CTAT} = \frac{V_T \ln(N)}{R_1} + \frac{V_{BE1}}{R_2} \quad (4.4)$$

V_T : Thermal voltage ($\approx 26mV$ at room temperature i.e. 300K).

N : The ratio of the BJTs.

is given as

$$V_O = I_{ref} \cdot R_4 = \frac{R_4}{R_2} \left(V_{BE1} + \frac{R_2}{R_1} \cdot V_T \ln(N) \right) \quad (4.5)$$

4.6.2 PMOS Error Amplifier

The error amplifier in the BGR is key to ensuring the proper operating point for the circuit. The error amplifier forces the necessary voltage dependencies at the inputs and the outputs of the BJTs and resistors to create the compensating PTAT and CTAT currents. For this particular design, a two-stage opamp with a PMOS differential input stage is used to provide sufficient gain and appropriate biasing when using low supply voltages.

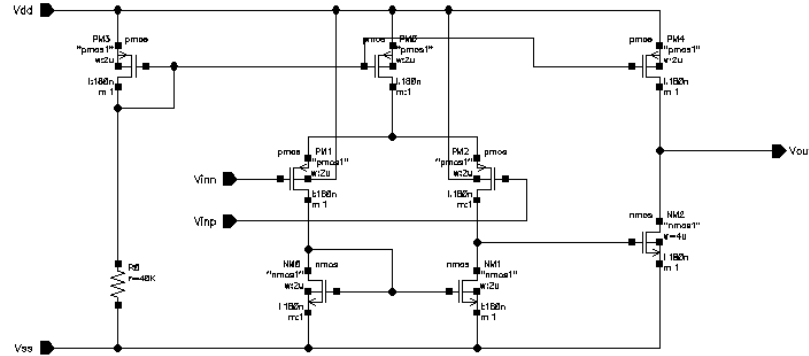


Figure 4.8: BGR Error Amplifier

In the first stage, we have a PMOS differential pair with a load made up of an NMOS current mirror. It should be noted that using PMOS devices as the input stage in the bandgap reference topology has its advantages because, by using PMOS devices, we ensure that the input voltage common-mode can operate at voltages very close to ground, which is usually characteristic for BGRs since the V_{BE} voltage level will also be very low.

Below are the Frequency Analysis and CMRR curve of the error amplifier.

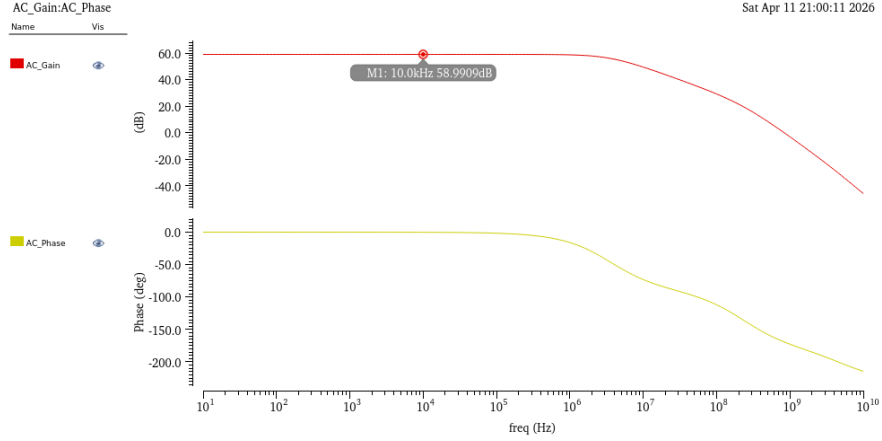


Figure 4.9: Frequency Response of BGR Error Amplifier

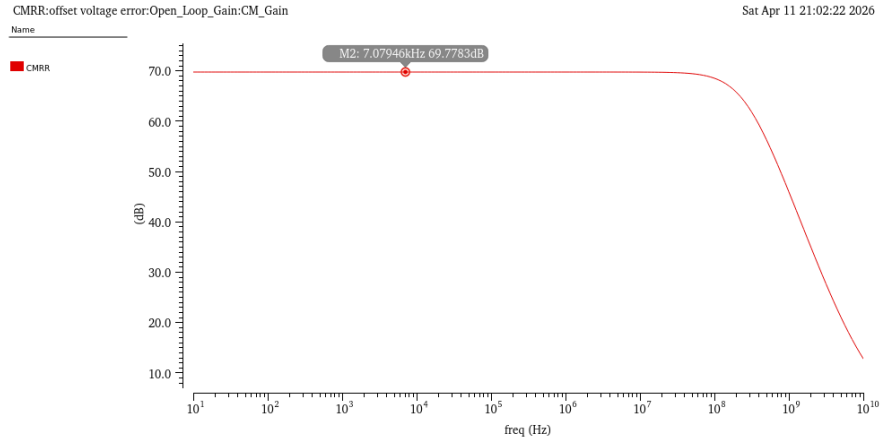


Figure 4.10: Common-Mode Rejection Ratio

4.6.3 Resistors Ratio

To achieve Zero T_C , the total change in current over temperature must be zero, hence upon partially differentiating the current equation with respect to the temperature we get the appropriate resistor ratio which can be fine tuned to achieve zero temperature dependence. This is given by

$$\frac{R_2}{R_1} = \frac{-(\partial V_{BE1}/\partial T)}{\ln(N) \cdot (\partial V_T/\partial T)} \quad (4.6)$$

Where

$$(\partial V_T/\partial T) = 0.0862 \text{ mV/K}$$

$$(\partial V_{BE1}/\partial T) \approx -1.8 \text{ mV/K}$$

$$N = 8$$

4.6.4 BGR Operation

The Temperature Independence has been achieved by Adding the CTAT and PTAT currents as shown in the figure below

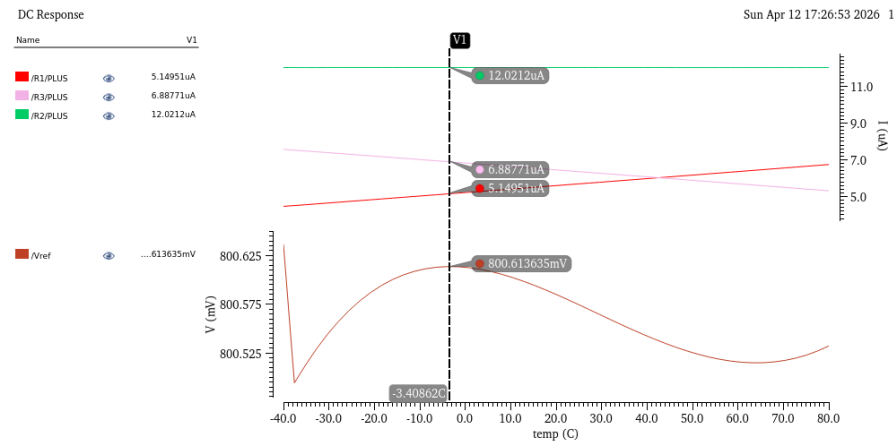


Figure 4.11: Temperature Sweep



System Architecture and Integration

5.1 Introduction

The various aspects of the proposed system components, such as the PV model, the machine learning based MPPT control, the charge pump, and the LDO, have been discussed in detail in the preceding chapters.

This chapter considers the design of the energy harvesting system by putting together all the various subsystems. This chapter will focus on the interactions among the subsystems, signal propagation in the system, and energy conversion within the system.

5.2 Overall System Architecture

The proposed system is essentially a multi-staged system design used to convert the input solar energy into regulated electrical energy. The main blocks of the system comprise of:

- Photovoltaic (PV) Cell
- Machine Learning-based MPPT Controller
- Voltage Controlled Oscillator (VCO)
- Charge Pump (Voltage Doubler)
- Programmable Low Dropout Regulator (LDO)
- Bandgap Reference Circuit (BGR)

Energy conversion is done in a sequential manner whereby the energy is first produced, optimized, boosted, and then regulated.

5.3 Functional Interconnection of Blocks

The PV cell serves as the main power source, creating both voltage and current depending on various external factors, like light irradiation level and temperature.

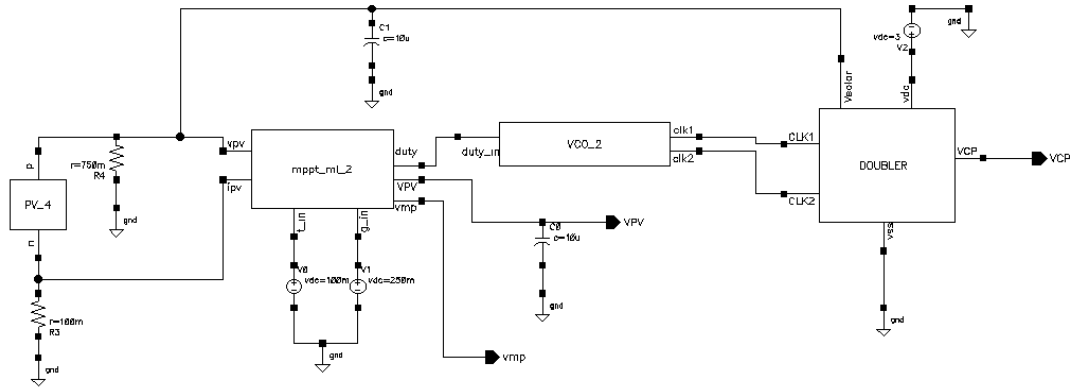


Figure 5.1: Integrated Solar Energy Harvesting System

The MPPT control takes the measured values of PV voltage and current, external environmental parameters into account, making predictions of the maximum power operating voltage point. A duty cycle is generated accordingly to this prediction.

The resulting duty cycle drives the Voltage Control Oscillator (VCO) to produce a series of clock signals with high frequency needed by the switching process in the charge pump module.

This charge pump module performs the boosting operation by means of capacitive switching. The resultant voltage is sent to the Low Dropout regulator and the BGR simultaneously.

The BGR produces a temperature independent and supply independent reference voltage V_{ref} as output which is then used by the programmable LDO for regulation.

Finally, the programmable LDO uses the reference voltage to produce an output voltage that remains fixed irrespective of ripples in the input value.

5.4 Signal Flow Analysis

The working of the proposed system can be seen in the signalling process within various blocks:

- The solar cell produces the signals (in terms of voltage and current)
- The MPPT controller processes those signals

- A control signal corresponding to the duty cycle produced by the MPPT controller
- Conversion of duty cycle to switching pulses via the VCO
- The charge pump employs those switching pulses for voltage boosting
- The BGR utilizes this boosted voltage for generating a reference voltage by combining CTAT and PTAT components of the circuit.
- The boosted voltage acts as the input to the LDO which generates a feedback voltage which is compared to the reference voltage.
- The LDO thus stabilizes the boosted voltage for final output

This signal flow ensures proper coordination between control and power stages, enabling efficient energy harvesting and conversion.

5.5 Control and Power Interaction

The system integrates both control and power processing mechanisms:

- The MPPT controller acts as the decision-making unit
- The VCO and charge pump act as energy processing units
- The LDO acts as the regulation unit

A very crucial role in coordinating the two stages is played by the duty cycle generated by the MPPT controller. Any changes in the environment will lead to a change in the duty cycle that will affect energy transfer.

5.6 Implementation Strategy

The implementation of the system is carried out via a behavioral/circuit-level approach:

- PV model and MPPT controller are implemented using Verilog-A
- VCO is modeled behaviorally for flexible control
- Charge pump, LDO and BGR are designed at the transistor level

Through this approach, both the system dynamics and the circuits are modeled. The entire system is modeled in Cadence Virtuoso Environment.

5.7 Design Considerations

During system integration, several design aspects are considered:

- Proper interfacing between analog and switching blocks
- Stability of control loop under varying conditions
- Efficient voltage boosting with minimal losses
- Compatibility between charge pump output and LDO input
- Proper regulation due to constant reference voltage

These considerations ensure reliable operation of the complete system.

5.8 Summary

In this chapter, we have looked at the development of the overall energy harvesting system by integrating all the sub-systems together. We have looked at how different components interact with each other and signal flow through the system.

The designed system acts as an integrated energy harvesting system pipeline, which can operate smoothly. The following chapter will concentrate on the results obtained by simulation.



Result and Analysis

6.1 Introduction

Simulation results for the overall energy harvester system will be discussed in this chapter. In contrast to earlier chapters where separate blocks were considered individually, this chapter looks at the overall behavior of the energy harvester system.

The purpose of this section is to determine how well the system manages energy harvesting, energy control, voltage conversion, and regulation when all blocks work together.

6.2 System-Level Transient Response

Transient behavior for the complete system was simulated using Cadence Virtuoso software.

From the result, it can be seen that the system attains a steady state after some time. This implies that the photovoltaic voltage is almost constant, meaning that the system works in a stable regime.

The MPPT controller's output, which comprises maximum power voltage and duty cycle, converges smoothly without oscillations.

This indicates that the feedback control loop is stable, hence, no instabilities are created by the control system.

6.3 MPPT Control Behavior in Integrated System

The duty cycle is produced from the MPPT controller by analyzing the environmental data and PV voltage.

From the obtained simulation results, it can be noted that:

- The duty cycle fluctuates at first and then reaches a steady value.
- The estimated maximum power point voltage stays within a bounded limit.

- PV voltage is maintained near a steady operating state.

It is clear that the designed controller successfully holds a steady operating point under steady state.

6.4 VCO and Switching Signal Analysis

The voltage-controlled oscillator takes input in the form of a duty cycle and produces clock signals used in the charge pump.

The simulation results show that:

- Two complementary clock pulses are produced.
- The output pulses are periodic and stable.
- The signal frequency stays constant after an initial transient period.

It shows that VCO successfully generates a reliable signal for the operation of the charge pump.

6.5 Charge Pump Output Performance

The charge pump stage is designed for voltage boosting using the voltage provided by the PV source.

From the results:

- The output voltage is much higher than the PV input voltage
- Boosting process stabilises from its initial transient state
- The output voltage ripple is within acceptable limits

Figure 6.1 shows the transient response of the charge pump, where the input voltage of approximately 2 V is boosted to around 3.27 V using complementary clock signals.

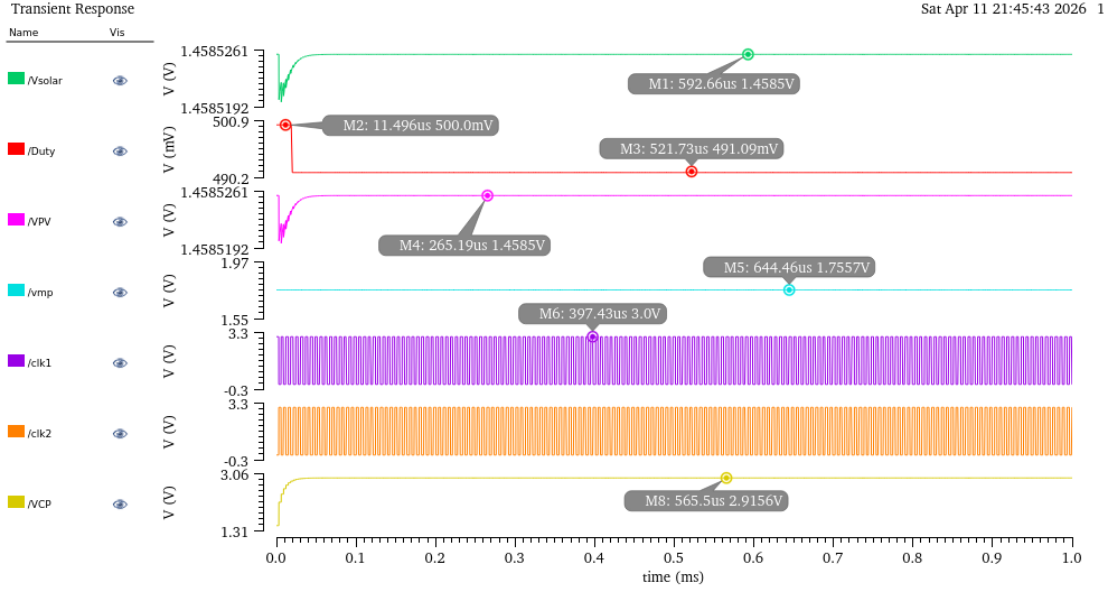


Figure 6.1: Output Waveforms of Integrated System

6.6 Overall System Energy Conversion

We have developed a multi-stage process of energy conversion:

- PV cell generates electrical energy
- MPPT controller determines control action
- VCO produces switching signals
- Charge pump boosts voltage

The cooperative functioning of these blocks leads to the formation of a full-fledged energy harvesting system.

Control stage and power conversion stage provide a stable operation of the system without any instabilities.

6.7 Error Amplifiers Performance

The performance of the error amplifier plays an important role in determining the functioning of the LDO since it directly impacts the accuracy of the regulation process, the stability of the circuit, and its transient behavior. Important parameters include the open loop gain, the bandwidth, which controls the speed

of the response; and the phase margin, which determines stability in feedback operation. Other parameters, such as slew rate also play a crucial role when it comes to driving the pass transistor properly. The table below lists all important parameters of the designed error amplifier.

Parameters	NMOS EA	PMOS EA
Gain	56.52 dB	58.91 dB
Bandwidth	6.419 MHz	3.52 MHz
Unity Gain Frequency	1.12 GHz	839.33 MHz
CMRR	53.20 dB	69.77 dB
Phase Margin	15.2 deg @ 787.87 MHz	18.2 deg @ 582.81 MHz
Gain Margin	10.89 dB @ 1.49 GHz	15.56 dB @ 1.46 GHz
Slew Rate	2.039 V/ μ s	1.907 V/ μ s
THD	0.304	0.360
PSRR	-67.73 dB	-73.44 dB

Table 6.1: Error Amplifiers performance metrics

6.8 LDO Transient Behaviour

- The transient behavior of the LDO will decide the time taken by the circuit for reaching the stable output voltage after the occurrence of any abrupt change in the load current or other parameters.
- In the present design, the output voltage can be controlled in digital mode with the help of control signals S_1S_0 , where the output voltages of 0V, 1.2V, 1.8V, and 2.5V can be selected from states 00, 01, 10, and 11, respectively.

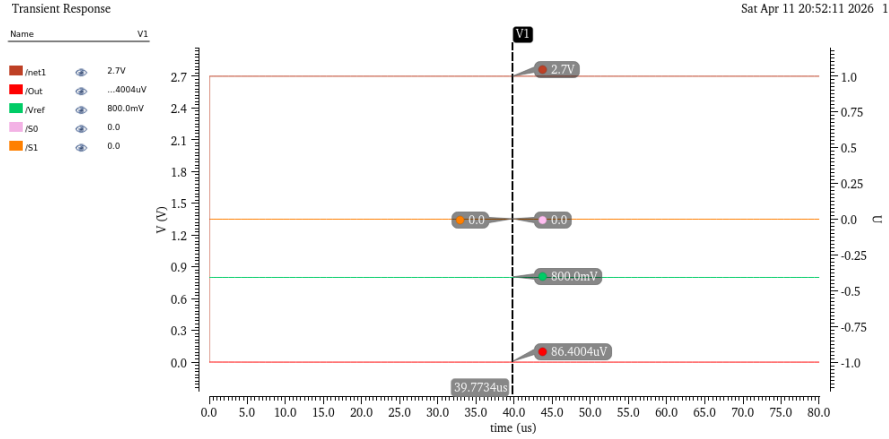


Figure 6.2: Transient Response when $S_1S_0 = 00$

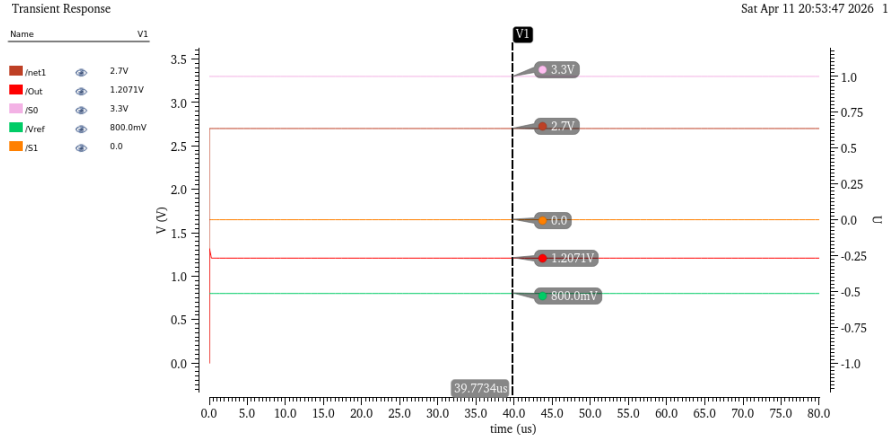


Figure 6.3: Transient Response when $S_1S_0 = 01$

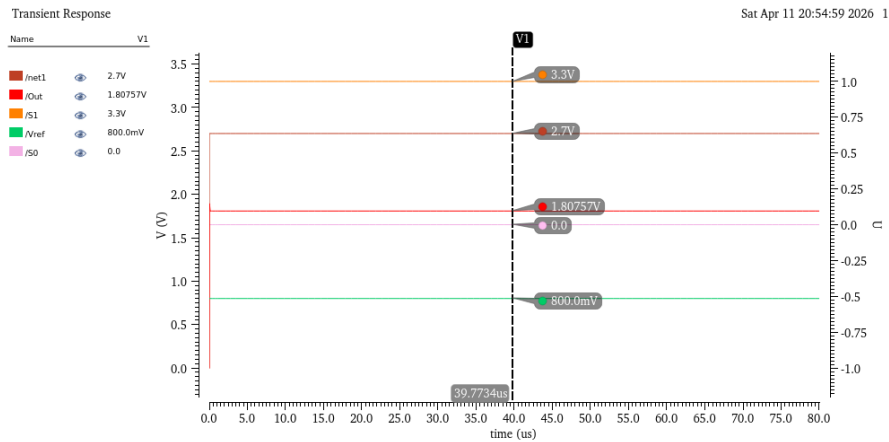


Figure 6.4: Transient Response when $S_1S_0 = 10$

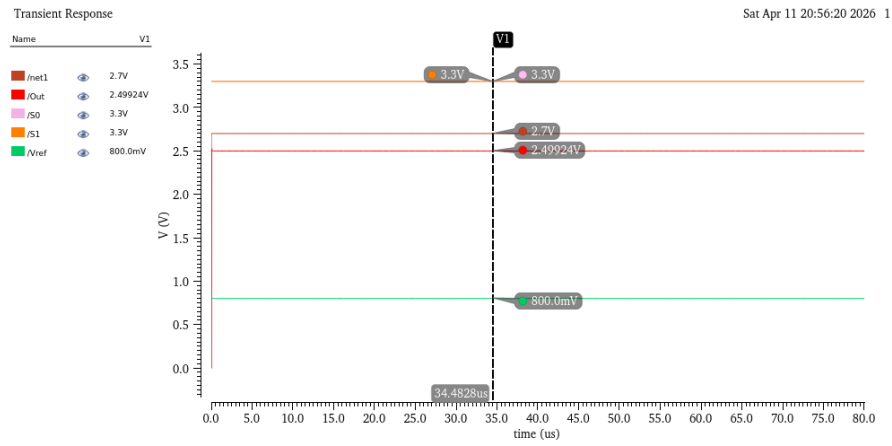


Figure 6.5: Transient Response when $S_1 S_0 = 11$

6.9 Discussion

Based on the results of our simulations, we can conclude that the proposed structure is capable of stable operation and energy conversion.

MPPT stage controls the stability of the system operation, whereas voltage boost stage provides voltage boosting needed for other stages of processing.

Despite the relatively minor changes in PV voltage at simulated values, the system provides closed-loop operation and successful interaction of all system parts.

6.10 Summary

This chapter has provided the system-level results of the energy harvesting architecture.

The obtained results prove the capability of our design to function stably and produce control signals and voltage boosting.



Conclusion

This thesis describes an integrated energy harvesting solution using a photovoltaic (PV) cell as a source of energy. The system incorporates machine learning MPPT controller, voltage-controlled oscillator (VCO), charge pump DC-DC converter, and low dropout regulator (LDO).

The nonlinear features and the dependency on irradiance and temperature of the PV cell have been considered during the modeling process. Machine learning MPPT control algorithm has been implemented that allows the prediction of the optimal voltage without having to use iterative MPPT approaches.

An MPPT controller outputs the value of duty cycle based on the maximum power point and current PV cell voltage. The value is used to regulate VCO. VCO creates switching signals necessary to regulate the charge pump operation. Charge pump allows converting low input voltage into high voltage. LDO regulates the resulting voltage.

All the subsystems have been proven correct independently and also in the context of the whole circuit when simulated in Cadence Virtuoso.

Thus, an architecture of an integrated energy harvester is proposed. It encompasses all the stages of energy harvesting starting from generation to voltage regulation.

Future Scope

Further improvement of the suggested technique may be achieved by making enhancements to the machine learning algorithm used in the MPPT process by applying new models that can increase the precision during fast changes in solar irradiance and temperature.

The charge pump efficiency may be increased by applying different methods of switching. Multi-stage versions may be considered for increasing voltage gain.

Implementation of the entire design on a layout level with experimental validation may be done in future research works. Further optimization of the LDO may be applied for achieving greater efficiency and transient response.

Finally, incorporation of an energy storage unit may be done, along with modifications for practical applications such as Internet of Things (IoT) and low-power embedded systems.

ANNEXURE A

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